

Agenda.

- how to approach Schema Design
- MVC Architecture
- Design TicTacToe

how to approach Schema Design

- Nouns
- Primitive attrs.
- Non-Primitive attrs.

Other entity / class attrs.

→ Cardinality.

- $1:1 \Rightarrow$ Id of one side on other side.
- $1:M / M:1 \Rightarrow$ Id of one side on M side.
- $M:M \Rightarrow$ Mapping / lookUp / Junction table

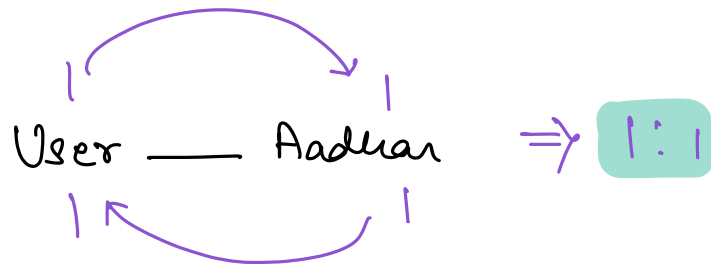
users

	aadhar-id

aadhar

	user-id

OR

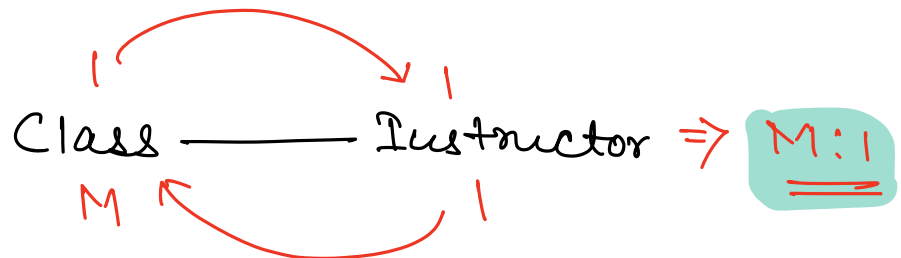


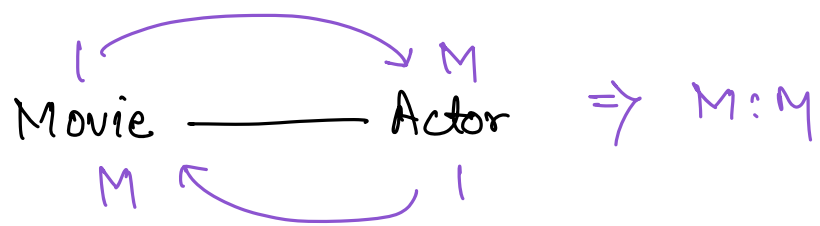
Classes

	instructor-id
1234	104

instructors

	classes
104 —	[...]





movies

	list of actors
	X

OR

actors

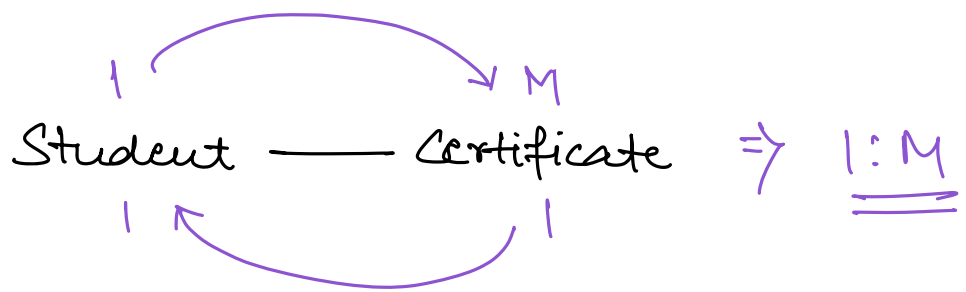
	list of movies
	X



movies - actors

movie_id	actor_id

Quiz

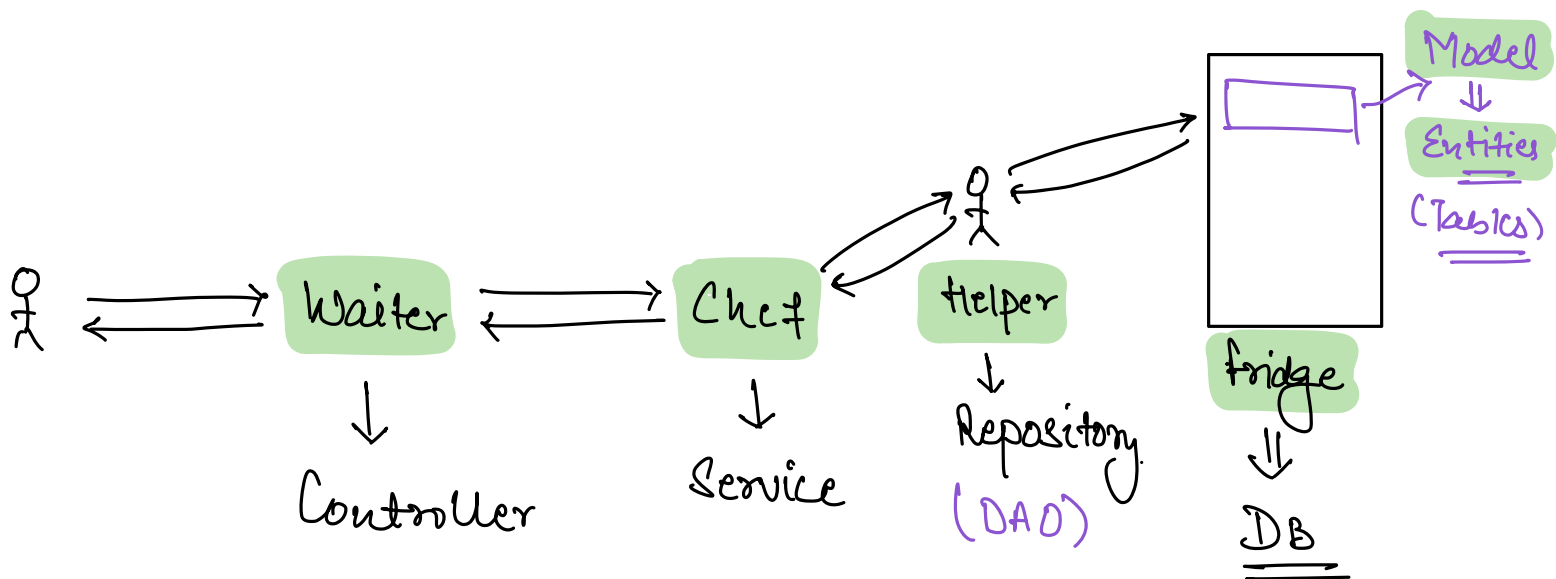


How to Code : MVC Architecture

Model

View

Controller.



BookMyShow

/controllers/

/services/

/repositories/

/models/

Design TicTacToe

- No need to persist data.
- Terminal based App.

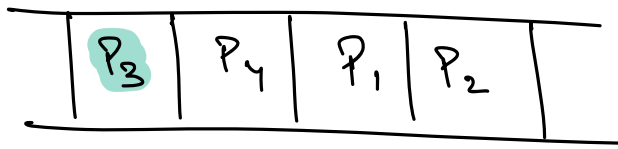
→ Overview ✓

- Requirements
- Clarification
- Class Diagram
- Schema Design
- Code.

Requirements.

- Size of the board is $N \times N$.
- No. of players = $N-1$
- Each player will have their own symbol.
- Every player can choose their symbol at the start and No two players should have same symbol.
- Will there be a Bot? Yes
- Only 1 bot is allowed per game.
- Bot can have different difficulty levels.
↓
EASY | MEDIUM | HARD.
- Game will end if:
 - Someone wins the game.
 - DRAWs.
- A player will WIN the game if they get their symbol across any row or any column or any diagonal.

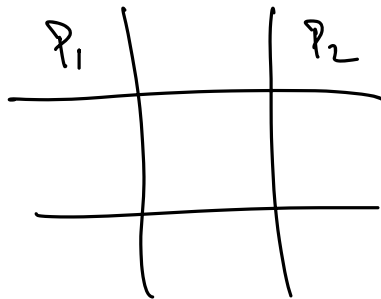
→ Who will start the game?



→ Undo.

↳ Global Undo.

$\text{list}(\text{Move}) = \left[\begin{Bmatrix} P_1 \\ 0 \\ 0 \end{Bmatrix}, \begin{Bmatrix} P_2 \\ 0 \\ 2 \end{Bmatrix}, \begin{Bmatrix} P_1 \\ 1 \\ 1 \end{Bmatrix}, \dots \right]$



→ Remove the last element from the list.

→ Make the cell empty on the board.

list of board state after each move

{

P_1		

 ,

P_1		
		P_2

 ,

P_1		
		P_2
	P_1	

 ,

P_1		
	P_2	P_2
	P_1	

 , ... }



Current
Board.

Instead of undoing the n^{th} turn, we can redo the first $(n-1)$ turns from scratch.

Class Diagram

